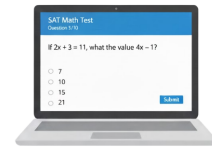


Does SAT Prep Improve Scores?



A certain high school offered an after-school SAT prep class that students could volunteer to take. 44 students took the course and then took the SAT. The average SAT score for this group was 1220. The average SAT score for all students who did not take the prep class was 1050.

1. Is the situation described an observational study or an experiment?
no treatment ← *treatment imposed*

Observational study, as the students are not forced to take the class.

2. Identify the explanatory variable and the response variable.

also called a factor ← Explanatory → ~~SAT prep class~~ whether or not took prep course
Response → SAT score

3. Can you conclude that taking the prep course will cause a student's SAT score to increase? Why or why not?

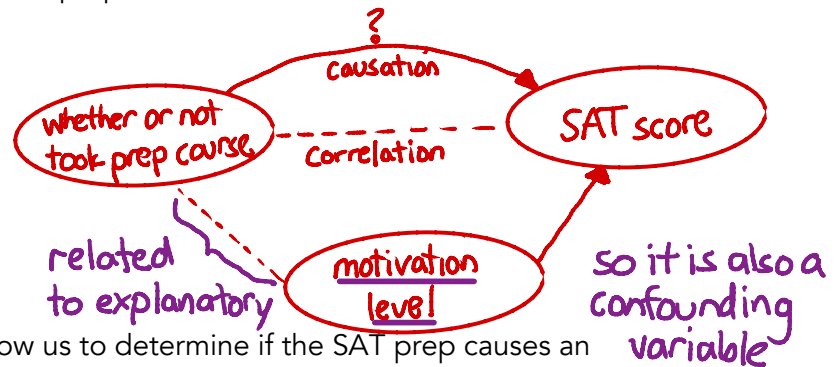
No, students who take the prep course might be highly motivated so they study more anyways.

need an experiment to show causation

4. Identify as many other possible variables that you can that may explain why the SAT scores are higher for those who took the prep course than for those who did not.

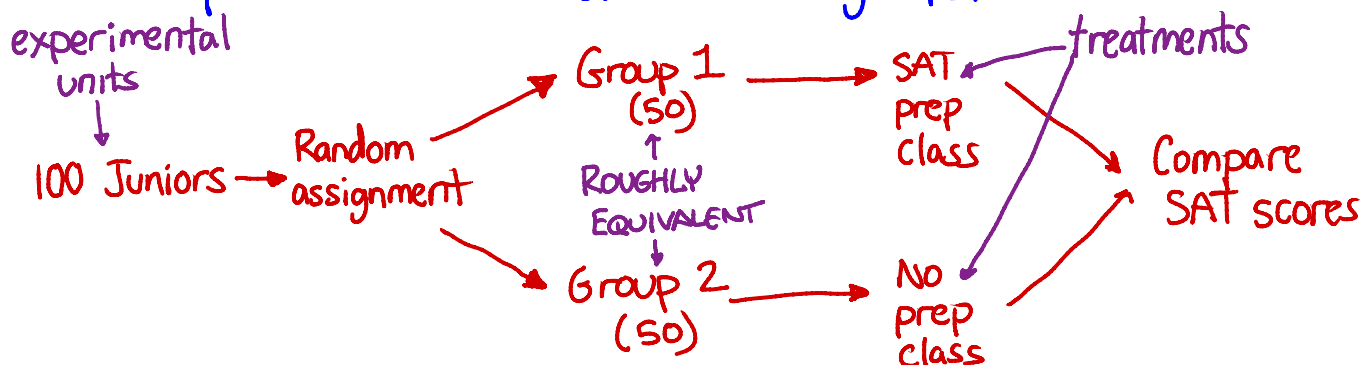
- motivation level
- college bound
- time after school
- number AP classes

extraneous variables (affect response variable)



5. Design an experiment that would allow us to determine if the SAT prep causes an increase in SAT scores. Be as thorough as possible.

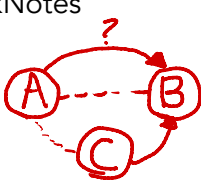
Randomly choose 100 Juniors to be part of the experiment. Randomly split the group so half take the prep course and half do not. One month later have all students take the SAT and compare scores between the two groups.



Lesson 2.5 – Observational Studies and Experiments

QuickNotes

LT#1



A → explanatory variable (factor) B → response variable

C → confounding variable - related to explanatory variable
- influences the response variable

LT#2

Observational study: no treatment imposed

Experiment: treatments imposed, allow us to show causation

LT#3

Experimental units: what/who treatment is imposed on.

Treatments - what is done (or not done) to experimental units
- levels or combination of levels of the explanatory variable(s)

Check Your Understanding

1. Do homes with metal roofs get worse cell service than homes with shingled roofs? Ben and Jerry are neighbors and own an ice cream business together. Ben has a metal roof on his home and Jerry has a shingled roof. They select a random sample of 50 employees and randomly assign them to come to a party at either Ben's house or Jerry's house. When they arrive Ben and Jerry ask them about their cell phone signal strength. Was this an observational study or an experiment? Justify your answer.

Experiment, because a treatment (Ben's house or Jerry's house) is being randomly assigned for the employees.

2. A study showed that one variable that strongly correlates with student academic success is the number of books that the family has in their home. Researchers randomly selected 100 students and measured their academic success as well as counted the number of books their family had in their home.
 - a. Is this an observational study or an experiment? Justify your answer.

Observational study because no treatment was imposed.

- b. What are the explanatory and response variables?

Explanatory: number of books in the home

Response: academic success

- c. Explain clearly why such a study cannot establish a cause-and-effect relationship. Suggest a variable that may be confounded with number of books that the family have in their home.

We can't make conclusions about a cause-and-effect relationship without an experiment. Maybe students with more books tend to study more, and studying results in greater academic success.